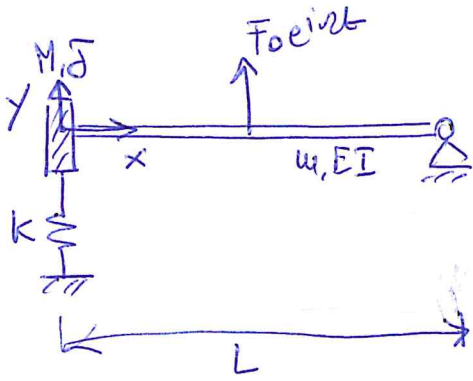
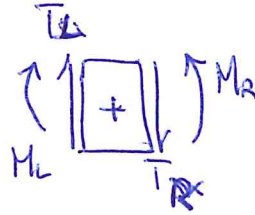


PROBLEM 2

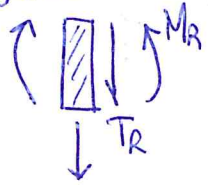


$$w(x,t) = (A \cos \gamma x + B \sin \gamma x + C \cosh \gamma x + D \sinh \gamma x) \cdot \cos(\omega t + \varphi)$$



$$\omega^2 = \gamma^4 \frac{EI}{m}$$

$$\delta \ddot{\theta}(0,t)$$



$$kW(0,t) + M\ddot{y}(0,t)$$

$$\bullet \sum F_y = 0 : kW(0,t) + M\ddot{y}(0,t) + EI w'''(0,t) = 0$$

$$k(A+C) - M\omega^2(A+C) + EI\gamma^3(-B+D) = 0$$

$$\left(k - M\gamma^4 \frac{EI}{m}\right)A - EI\gamma^3 B + \left(k - M\gamma^4 \frac{EI}{m}\right)C + EI\gamma^3 D = 0$$

$$\bullet \sum M = 0 : \delta \ddot{\theta}(0,t) - M_R = 0$$

$$(\ddot{\theta} = \frac{\partial^2 \theta}{\partial t^2} ; \vartheta = \frac{\partial w}{\partial x})$$

$$- \delta \omega^2 (B\gamma + D\gamma) - EI\gamma^2 (-A+C) = 0$$

$$EI\gamma^2 A - \delta \gamma^5 \frac{EI}{m} B - EI\gamma^2 C - \delta \gamma^5 \frac{EI}{m} D = 0$$

$$\bullet w(L,t) = 0 : A \cos \gamma L + B \sin \gamma L + C \cosh \gamma L + D \sinh \gamma L = 0$$

$$\bullet w'(L,t) = 0 : EI(-A\gamma^2 \cos \gamma L - B\gamma^2 \sin \gamma L + C\gamma^2 \cosh \gamma L + D\gamma^2 \sinh \gamma L) = 0$$

$$[H(\gamma)] = \begin{bmatrix} \alpha & -EI\gamma^3 & \alpha & EI\gamma^3 \\ 1 & -\frac{\delta \gamma^3}{m} & -1 & -\frac{\delta \gamma^3}{m} \\ \cos \gamma L & \sin \gamma L & \cosh \gamma L & \sinh \gamma L \\ -\cos \gamma L & -\sin \gamma L & \cosh \gamma L & \sinh \gamma L \end{bmatrix}$$

$$\alpha = k - M\gamma^4 \frac{EI}{m}$$

$$k_{mi} = k_{mi,beam} + k \phi_i^2(\varphi)$$

$$m_{mi} = m_{mi,beam} + M \phi_i^2(\varphi) + \delta \phi_i^2(\varphi)$$

$$w(x,t) = \underline{\phi}^T(x) \underline{q}(t)$$